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SUPERSEDING
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MILITARY SPECIFICATION

# CLOTH, TOW TARGET, ACETATE AND RAYON

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the requirements for a rip-stop-twill weave acetate and rayon cloth, intended for use in the manufacture of aerial tow targets.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| CCC-T-191 | Textile Test Methods |
| PPP-P-51 | Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics) |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |

FSC 8305

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publication - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D. C. 20580.)

## 3. REQUIREMENTS

3.1 Material - The rip-stop-twill weave cloth shall be calendered and made from a combination of bright acetate yarns and medium high-tenacity viscose rayon yarns. The rip-stop ends and picks shall be of sufficient tenacity so that the finished product conforms to this specification.

3.2 Weave - The basic weave shall be a 2 up and 1 down twill with two warp ends weaving as one throughout and shall be as shown in Figure 1. The yarns shall repeat on 31 ends of the bright acetate and 3 ends of medium high-tenacity viscose rayon and filling on 16 picks of bright acetate and 2 picks of medium high-tenacity viscose rayon.

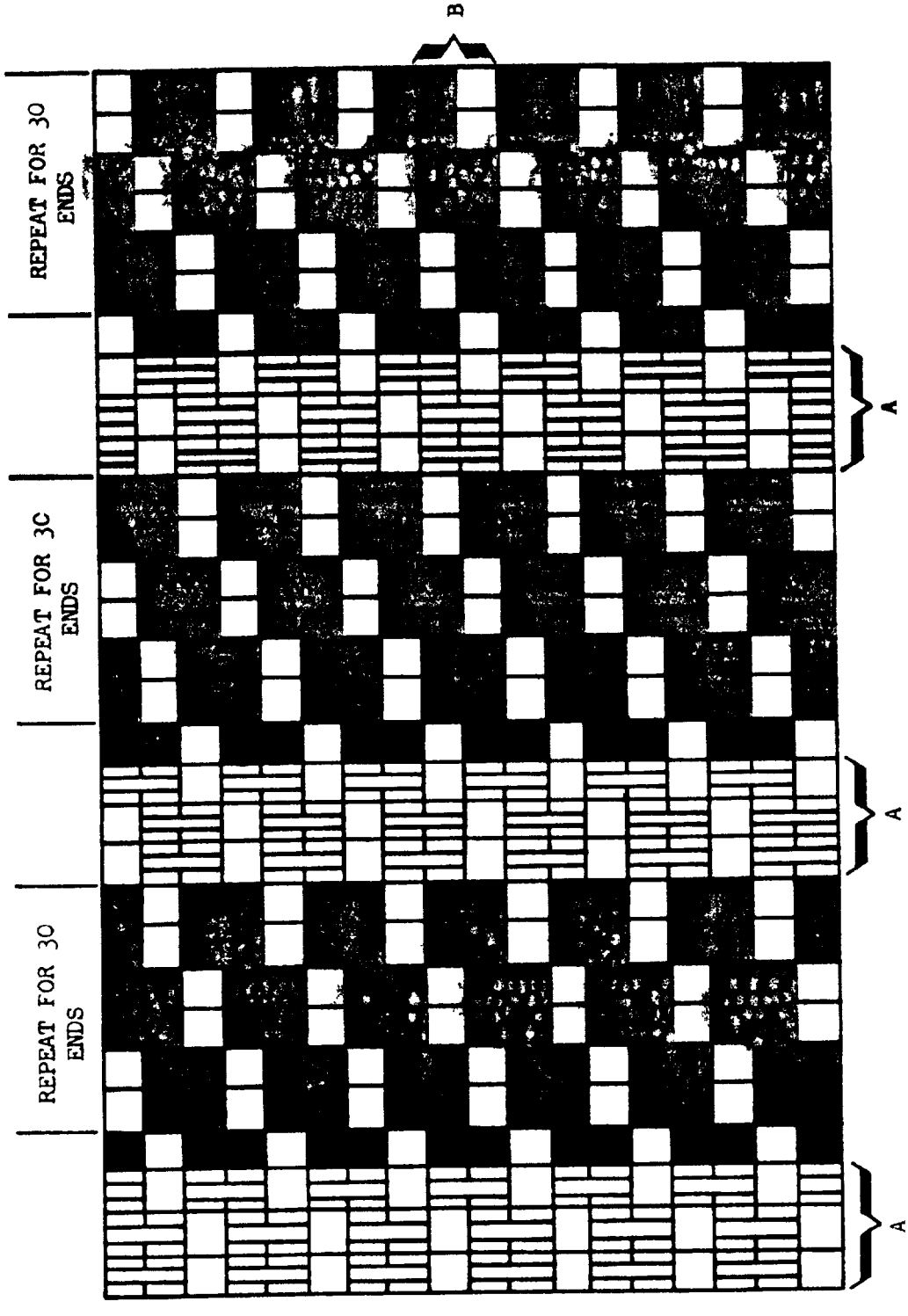
3.3 Physical properties - The finished cloth shall conform to the requirements specified in Table 1.

3.4 Color - The color of the finished cloth shall be as specified by the procuring activity (see 6.2).

3.5 Nonfibrous material - The chloroform-extractable material shall be not more than 2.0 percent when tested as specified in Table V.

3.6 Finish - The cloth shall be scoured and shall contain no sizing, lubrication, or weighting material. The face of the cloth shall present a smooth, glazed surface after finishing.

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A = VISCOSE RAYON WARP ENDS
B = VISCOSE RAYON FILLING PICKS

FIGURE 1. Weave diagram



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* TABLE I - PHYSICAL PROPERTIES

| CHARACTERISTIC | REQUIREMENT |
|---|---|
| Weight, ounces per square yard, maximum | 7.5 |
| Yarns per inch, minimum | |
| Warp | 14 |
| Filling | 58 |
| Breaking strength, pounds, minimum | |
| Warp | 65 |
| Filling | 50 |
| Elongation, percent, maximum | |
| Warp | 25 |
| Filling | 25 |
| Tearing strength, pounds, minimum | |
| Warp | 5 |
| Filling | 5 |
| Colorfastness to light (20 standard fading hours) | Good |
| Colorfastness to water, salt water without soap, and crocking | Good |
| Acidity (pH) | 6.0 to 8.0 |
| Air permeability, cu. ft. air/min./ft.$^2$ at 1/2 inch water pressure | 1.5 -- 5.0 |

* 3.7 Fiber identification - Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

* 3.8 Length and put-up - Unless otherwise specified (see 6.2), the cloth shall be in continuous lengths, each not less than 40 yards. The lengths shall be put up on rolls as specified in PPP-P-51.



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* 3.9 Width - The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

* 3.10 Workmanship - The finished cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

* 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance - Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification. (See Table V).

4.2 Sampling and inspection for acceptance - Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Inspection of the end item - The end item shall be inspected for defects in accordance with classifications in 4.2.1.1, 4.2.1.2, and 4.2.1.3.

4.2.1.1 Yard-by-yard examination - The required yardage of each roll shall be examined and the visual defects classified as listed in Table II. All defects found shall be counted regardless of their proximity, one to another, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The sample size shall be in accordance with Inspection Level II of MIL-STD-105. The acceptable quality level shall be 6.5 defects per 100 units (yards). The lot size shall be expressed in units of one (1) yard each. An approximate equal number of yards shall be examined from each roll selected. The number of rolls from which the sample is to be selected shall be in accordance with Table III.



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TABLE II - LIST OF DEFECTS

| DEFECTS |
|---|
| Broken or missing yarns - two contiguous, 2 inches or more in length. |
| Hole, cut, tear, any. |
| Spot, stain or streak more than 2 inches in combined directions. |
| Selvage, cut, torn, tight, rolled or loopy. |
| Open, thin, abraded or weak place. |
| Foreign or waste fiber, more than 1 inch in combined directions. |

TABLE III - LOT SIZE FOR ACCEPTANCE

| Lot size in yards | Sample size in rolls | Acceptance number |
|---|---|---|
| Up to 1300 | 3 | 0 |
| 1301 to 3200 | 5 | 0 |
| 3201 to 8000 | 7 | 0 |
| 8001 to 22,000 | 10 | 0 |
| 22,001 to 110,000 | 15 | 0 |
| 110,001 and up | 25 | 1 |

4.2.1.2 Overall examination - Each defect listed below shall be counted not more than once in each roll examined. The unit of product shall be one roll. The sample size or number of rolls selected as samples, and the maximum number of defects acceptable, shall be as shown in Table III.



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TABLE IV - OVERALL DEFECTS

| |
|---|
| Overall uncleanness, objectionable odor |
| Uneven weaving for more than 20 yards 1/ |
| Selvage curled, folded, rolled, slack, or loopy, over more than half the piece 1/ |
| Baggy, ridgy, or wavy cloth |
| Width, less than minimum specified |

1/ Intermittently or continuously

4.2.1.3 Length examination -

4.2.1.3.1 Examination for length of individual rolls - The required rolls shall be examined for the following defects: gross length found to be more than 2 yards below the gross lengths marked on the ticket, any piece less than 40 yards. The unit of product for this examination shall be one roll. The sample size and acceptance number shall be as shown in Table III. The number of rolls from which the sample is selected shall be in accordance with Table III.

4.2.1.3.2 Examination for total yardage in samples - The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets.

4.2.2 Examination of preparation for delivery requirements - An examination shall be made in accordance with the provisions of PPP-P-51 to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with.

4.3 Testing of the end item - The methods of testing specified in CCC-T-191, wherever applicable, and as listed in Table V shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The lot size shall be expressed in units of 1 linear yard. The unit of product for test purposes shall be 3 linear yards, full width of the finished cloth. The sample size shall be as listed in Table VI. The lot shall be unacceptable if any unit fails to meet any requirement specified.

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TABLE V - TEST METHODS

| Characteristic | Test method |
|---|---|
| Material 1/ | |
| Identification of viscose rayon | 1500 |
| Identification of acetate rayon | 1510 |
| Finish 1/ | |
| Weight, oz. per sq. yd. | 5041 |
| Yarns per inch | 5050 |
| Breaking strength, ravel strip, pounds 2/ | 5104 |
| Elongation, percent 2/ | 5104 |
| Tearing strength, tongue, pounds | 5134 |
| Colorfastness to: | |
| Light, 20 standard fading hours | 5660 |
| Water | 5630 |
| Salt water, without soap | 5632 |
| Nonfibrous material | |
| Chloroform soluble | 2611 |
| Crocking | 5650 |
| Acidity, pH | 2811 |
| Air permeability, cu. ft. air/min./ft. $^2$ at 1/2 inch water pressure | 5450 |

1/ A supplier's certificate of compliance is acceptable for this characteristic.

2/ Except that the face of both jaws of each clamp on the testing machine shall measure 1 inch by 1-1/2 inches or more with the long dimension perpendicular to the direction of load application.



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TABLE VI - SAMPLING FOR TEST

| Lot size (yards) | Sample size |
|---|---|
| 800 or less | 2 |
| 801 to 22,000 | 3 |
| 22,001 and over | 5 |

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking - The cloth shall be packaged, packed, and marked in accordance with PPP-P-51, Class 2. The levels of packaging and packing to be used in the preparation for delivery shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use - This cloth is intended for use in the fabrication of aerial tow targets.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Color required (see 3.4)

(c) Quantity and dimensions of cloth required (see 3.8 and 3.9)

(d) Selection of applicable levels of packaging and packing (see 5.1).

6.2.1 In the preparation of contracts and orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-51, may be exercised in the preparation for delivery of cloth.



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6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
Navy - AS

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Review activities:
Navy - AS
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*(See Instructions – Reverse Side)*

| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b ADDRESS *(Street, City, State ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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